4. A class investigated radioactive decay using 50 dice, each with 8 sides.

They rolled the dice and removed any which landed with an 8 facing upwards, to represent a decayed nucleus.

They recorded the number of dice remaining.

They rolled the remaining dice and again removed any which landed with an 8 facing upwards.

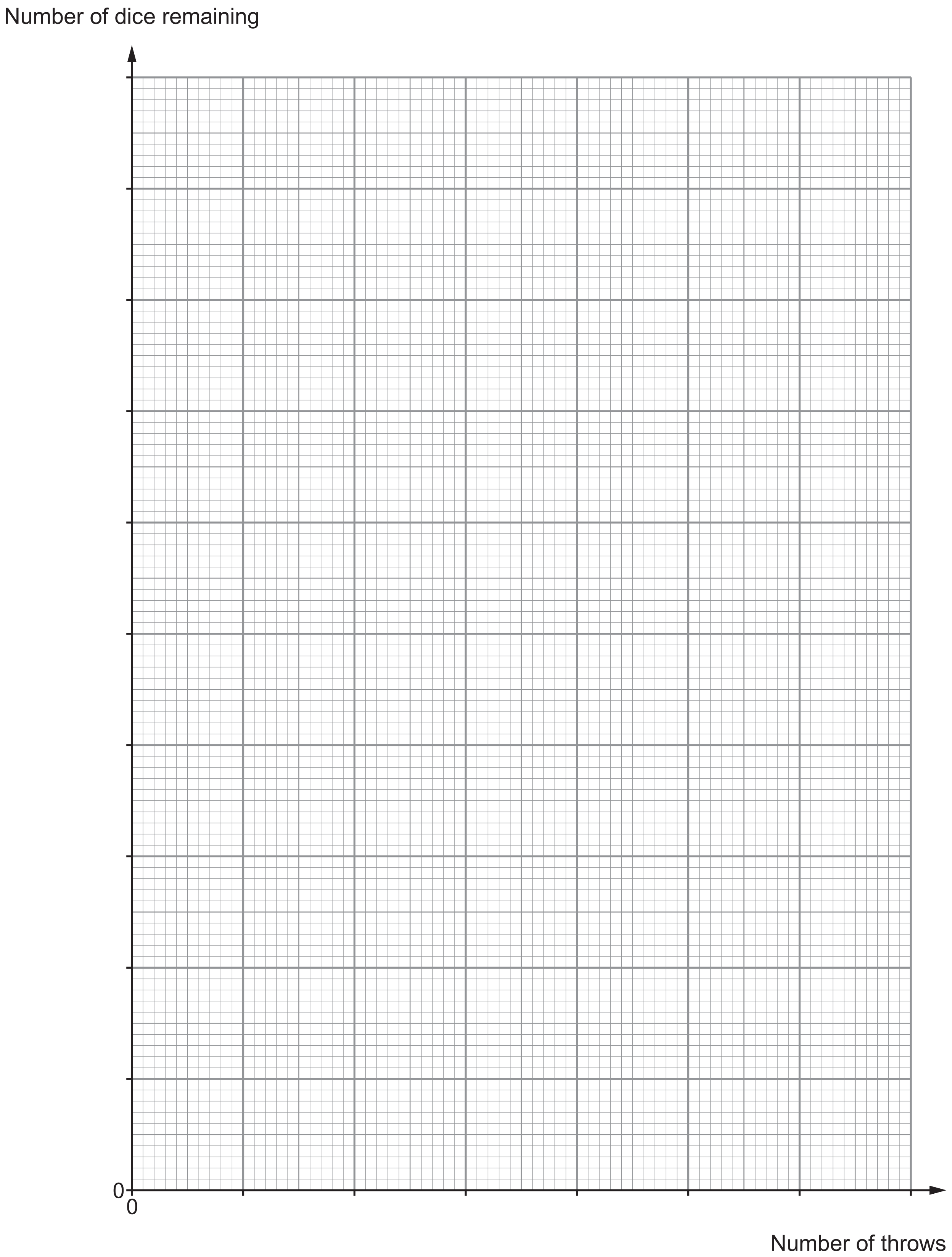
This was repeated until they had rolled the dice 10 times in total.

Some of the results are shown in the table below.

Number of throws	Number of dice remaining
0	50
2	37
4	29
6	23
8	18
10	15

(a) (i) Plot the data on the grid opposite and draw a suitable line. [3]

Examiner only



Examiner only

(ii) Use your graph to determine how many dice were removed on the 3^{rd} **throw**. [2]

Number of dice removed = ...

(iii) Determine how many throws it took for the initial number of dice to halve. Show how you obtained your answer on the graph. [2]

Number of throws = ...

(iv) If the experiment was repeated with 1 000 dice, use your results to predict how many throws it would take to reduce them to 125. [2]

Number of throws = ...

(b) Explain why it is preferable to use a larger sample size in this experiment. [2]

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